Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double Cross Dairy #4

Physical address of dairy:

Number and Street	City	County	Zip Code
6006 S Central AVE	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 03/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X057-X009-X006-XXXX X057-X009-X007-XXXX

B. OPERATORS

Ahlem, Mark

Operator name: Ahlem, Mark

Telephone no.: (209) 667-0667 Landline (209) 495-9221 Cellular

Mailing Address Number and Street	City	State	Zip Code
23546 W American AVE	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Chaves, Callie

Legal owner name: Chaves, Callie

Telephone no.: (209) 634-2823 Landline Cellular

Mailing Address Number and Street	City	State	Zip Code
2675 Nicklaus	Turlock	CA	95380

Chaves, Dave

Legal owner name: Chaves, Dave

Telephone no.: (209) 667-6166 Landline (209) 483-1825 Cellular

Mailing Address Number and Street	City	State	Zip Code
6100 S Central AVE	Turlock	CA	95380

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	150	0	0	0
Maximum number	0	0	200	0	0	0
Average number	0	0	150	0	0	0
Avg live weight (lbs)	0	0	800	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1,469.48 *tons per reporting period*

Total nitrogen from manure: 14,235.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 9,964.50 *lbs per reporting period*

Total phosphorus from manure: 2,409.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 14,049,000 *gallons*

Total nitrogen generated: 41,151.83 *lbs*

Total phosphorus generated: 5,717.66 *lbs*

Total potassium generated: 82,996.71 *lbs*

Total salt generated: 596,867.95 *lbs*

	24,975,000	*gallons applied*
+	0	*gallons exported*
-	10,926,000	*gallons imported*
=	14,049,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
01/10/2023	Process wastewater *WW*	510,000.00 *gal*	560.00	70.80	776.00		5,520
01/15/2023	Process wastewater *WW*	540,000.00 *gal*	560.00	70.80	776.00		5,520
02/09/2023	Process wastewater *WW*	390,000.00 *gal*	560.00	70.80	776.00		5,520
02/20/2023	Process wastewater *WW*	240,000.00 *gal*	560.00	70.80	776.00		5,520
02/28/2023	Process wastewater *WW*	405,000.00 *gal*	560.00	70.80	776.00		5,520
03/20/2023	Process wastewater *WW*	180,000.00 *gal*	560.00	70.80	776.00		5,520
04/26/2023	Process wastewater *WW*	420,000.00 *gal*	560.00	62.50	776.00		5,520
07/21/2023	Process wastewater *WW*	1,755,000.00 *gal*	171.00	62.50	714.00		3,540
08/02/2023	Process wastewater *WW*	1,050,000.00 *gal*	171.00	62.50	714.00		3,540
08/13/2023	Process wastewater *WW*	2,370,000.00 *gal*	171.00	62.50	714.00		3,540
08/25/2023	Process wastewater *WW*	990,000.00 *gal*	171.00	62.50	714.00		3,540
09/06/2023	Process wastewater *WW*	2,076,000.00 *gal*	171.00	62.50	714.00		3,540

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	24,307.41	5,855.47	66,489.91	367,132.77
Total imports for all materials	24,307.41	5,855.47	66,489.91	367,132.77

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
DC #1	48	48	2	process wastewater	X057-X080-X001-XXXX X057-X090-X007-XXXX
DC #2	19	19	2	process wastewater	X057-X080-X002-XXXX
DC #3	58	58	2	process wastewater	X057-X090-X006-XXXX
Totals for areas that were used for application	125	125	6		
Totals for areas that were not used for application					
Land application area totals	125	125	6		

B. CROPS AND HARVESTS

DC #1

Field name: DC #1

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 48 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	989.18 *ton*	As-is		71.6	5,200.00	1,100.00	7,400.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.61	214.32	45.34	305.00	1,533.39

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,337.20 *ton*	As-is		66.2	4,300.00	600.00	5,400.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.86	239.58	33.43	300.87	1,337.09

DC #2

Field name: DC #2

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	391.55 *ton*	As-is		64.0	5,500.00	1,200.00	8,800.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.61	226.69	49.46	362.70	1,676.66

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	541.82 *ton*	As-is		66.4	4,500.00	500.00	5,900.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.52	256.65	28.52	336.50	1,437.25

DC #3

Field name: DC #3

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 58 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	1,195.26 *ton*	As-is		74.4	4,200.00	1,000.00	7,100.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.61	173.11	41.22	292.63	1,424.42

DC #3

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,653.99 *ton*	As-is		68.8	4,300.00	600.00	5,800.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.52	245.25	34.22	330.80	1,423.57

NUTRIENT BUDGET

A. LAND APPLICATIONS

DC #1 - 11/15/2022: Oats, silage-soft dough

Field name: DC #1

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.11	3.00	59.46	707.24	600,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	292.91	8,022,960.00 *gal*
Application event totals		51.11	3.00	59.46	1,000.15	

01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	52.57	6.65	72.85	518.22	540,000.00 *gal*
Application event totals		52.57	6.65	72.85	518.22	

02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	37.97	4.80	52.62	374.27	390,000.00 *gal*
Application event totals		37.97	4.80	52.62	374.27	

03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.65	5.78	62.59	428.93	480,000.00 *gal*
Application event totals		47.65	5.78	62.59	428.93	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.75	1.77	35.47	307.10	240,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	212.13	5,810,400.00 *gal*
Application event totals		28.75	1.77	35.47	519.23	

DC #1 - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	208.09	5,699,568.00 *gal*
Application event totals		0.00	0.00	0.00	208.09	

DC #1 - 06/24/2023: Corn, silage

Field name: DC #1

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.18	7.54	150.73	1,305.16	1,020,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	260.97	7,147,920.00 *gal*
Application event totals		122.18	7.54	150.73	1,566.12	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	19.62	7.17	81.93	406.19	660,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	231.59	6,343,200.00 *gal*
Application event totals		19.62	7.17	81.93	637.78	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	31.22	11.41	130.34	646.22	1,050,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	215.57	5,904,480.00 *gal*
Application event totals		31.22	11.41	130.34	861.78	

DC #1 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	33.89	12.39	141.51	701.61	1,140,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	208.18	5,702,160.00 *gal*
Application event totals		33.89	12.39	141.51	909.79	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.30	10.13	132.59	780.88	1,140,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	212.69	5,825,568.00 *gal*
Application event totals		33.30	10.13	132.59	993.57	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.32	11.08	126.61	627.75	1,020,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	210.45	5,764,224.00 *gal*
Application event totals		30.32	11.08	126.61	838.20	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.29	8.00	104.68	616.49	900,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	201.16	5,509,824.00 *gal*
Application event totals		26.29	8.00	104.68	817.65	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	206.61	5,659,104.00 *gal*
Application event totals		0.00	0.00	0.00	206.61	

DC #2 - 11/15/2022: Oats, silage-soft dough

DC #2 - 11/15/2022: Oats, silage-soft dough

Field name: DC #2

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.83	1.52	30.04	357.34	120,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	273.47	2,964,950.00 *gal*
Application event totals		25.83	1.52	30.04	630.81	

01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.13	5.11	55.34	379.27	168,000.00 *gal*
Application event totals		42.13	5.11	55.34	379.27	

02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	59.03	7.46	81.80	581.87	240,000.00 *gal*
Application event totals		59.03	7.46	81.80	581.87	

03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	44.27	5.60	61.35	436.40	180,000.00 *gal*
Application event totals		44.27	5.60	61.35	436.40	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.05	1.79	35.84	310.33	96,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	225.64	2,446,402.00 *gal*
Application event totals		29.05	1.79	35.84	535.97	

DC #2 - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	29.51	3.73	40.90	290.93	120,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	205.76	2,230,885.00 *gal*
Application event totals		29.51	3.73	40.90	496.70	

DC #2 - 06/24/2023: Corn, silage

Field name: DC #2

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	264.27	2,865,200.00 *gal*
Application event totals		0.00	0.00	0.00	264.27	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	23.66	8.65	98.78	489.76	315,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	229.35	2,486,625.00 *gal*
Application event totals		23.66	8.65	98.78	719.12	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.90	24.91	326.15	1,920.84	1,110,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	214.51	2,325,695.00 *gal*
Application event totals		81.90	24.91	326.15	2,135.35	

DC #2 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.84	21.55	282.08	1,661.27	960,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	212.04	2,298,962.00 *gal*
Application event totals		70.84	21.55	282.08	1,873.31	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	74.35	27.18	310.46	1,539.26	990,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	213.88	2,318,836.00 *gal*
Application event totals		74.35	27.18	310.46	1,753.13	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.47	15.35	200.98	1,183.65	684,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	206.00	2,233,412.00 *gal*
Application event totals		50.47	15.35	200.98	1,389.65	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	206.89	2,243,121.00 *gal*
Application event totals		0.00	0.00	0.00	206.89	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	210.35	2,280,570.00 *gal*
Application event totals		0.00	0.00	0.00	210.35	

DC #3 - 11/15/2022: Oats, silage-soft dough

Field name: DC #3

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

DC #3 - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.30	2.49	49.21	585.30	600,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	300.37	9,941,200.00 *gal*
Application event totals		42.30	2.49	49.21	885.67	

01/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	41.09	5.20	56.94	405.05	510,000.00 *gal*
Application event totals		41.09	5.20	56.94	405.05	

02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.41	5.03	54.39	372.73	504,000.00 *gal*
Application event totals		41.41	5.03	54.39	372.73	

02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	32.63	4.13	45.22	321.66	405,000.00 *gal*
Application event totals		32.63	4.13	45.22	321.66	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.83	2.64	52.83	457.47	432,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	225.02	7,447,432.00 *gal*
Application event totals		42.83	2.64	52.83	682.49	

DC #3 - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	24.17	3.06	33.50	238.26	300,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	203.11	6,722,084.00 *gal*
Application event totals		24.17	3.06	33.50	441.37	

DC #3 - 06/24/2023: Corn, silage

Field name: DC #3

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.75	8.81	176.11	1,524.89	1,440,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	284.80	9,425,870.00 *gal*
Application event totals		142.75	8.81	176.11	1,809.69	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	19.19	7.01	80.13	397.28	780,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	245.56	8,127,076.00 *gal*
Application event totals		19.19	7.01	80.13	642.84	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.91	9.70	127.06	748.29	1,320,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	226.95	7,511,290.00 *gal*
Application event totals		31.91	9.70	127.06	975.24	

DC #3 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	30.26	11.06	126.36	626.48	1,230,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	212.18	7,022,466.00 *gal*
Application event totals		30.26	11.06	126.36	838.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.92	8.49	111.17	654.75	1,155,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	207.52	6,868,070.00 *gal*
Application event totals		27.92	8.49	111.17	862.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	25.98	9.50	108.48	537.86	1,056,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	210.50	6,966,786.00 *gal*
Application event totals		25.98	9.50	108.48	748.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.11	7.94	103.96	612.24	1,080,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	204.82	6,778,750.00 *gal*
Application event totals		26.11	7.94	103.96	817.05	

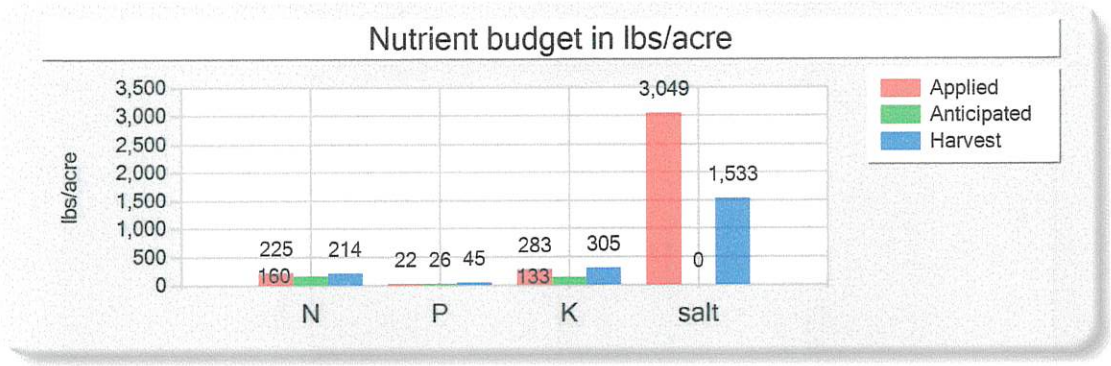
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	209.01	6,917,428.00 *gal*
Application event totals		0.00	0.00	0.00	209.01	

B. NUTRIENT BUDGET

DC #1 - 11/15/2022: Oats, silage-soft dough

Field name: DC #1 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.05	22.01	282.98	2,335.77
Fresh water	0.00	0.00	0.00	713.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.05	22.01	282.98	3,048.90
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	214.32	45.34	305.00	1,533.39
Nutrient balance	10.73	-23.33	-22.02	1,515.51
Applied to removed ratio	1.05	0.49	0.93	1.99

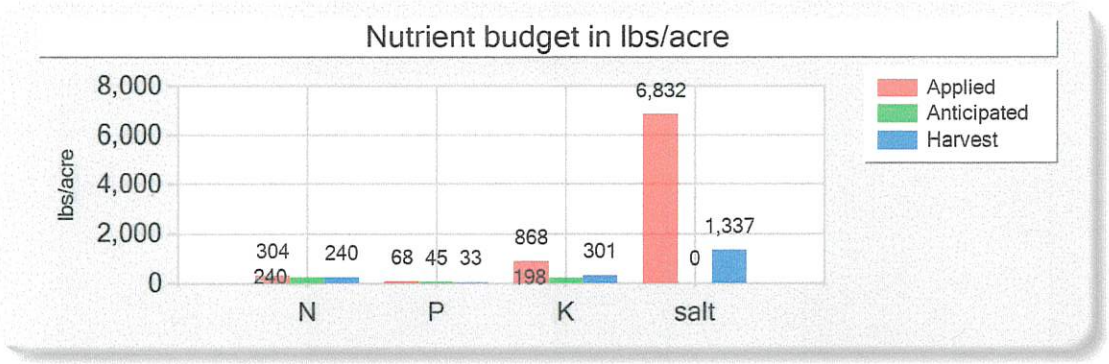
Fresh water applied	
19,532,928.00	*gallons*
719.33	*acre-inches*
14.99	*inches/acre*

Process wastewater applied	
2,250,000.00	*gallons*
82.86	*acre-inches*
1.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC #1 - 06/24/2023: Corn, silage

Field name: DC #1 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	296.82	67.71	868.39	5,084.30
Fresh water	0.00	0.00	0.00	1,747.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	303.82	67.71	868.39	6,831.51
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.58	33.43	300.87	1,337.09
Nutrient balance	64.23	34.28	567.52	5,494.42
Applied to removed ratio	1.27	2.03	2.89	5.11

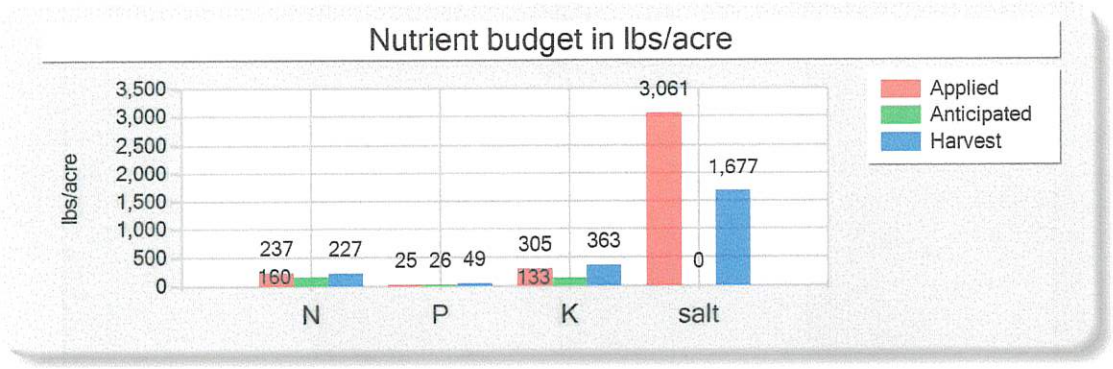
Fresh water applied
47,856,480.00 *gallons*
1,762.39 *acre-inches*
36.72 *inches/acre*

Process wastewater applied
6,930,000.00 *gallons*
255.21 *acre-inches*
5.32 *inches/acre*

Total harvests for the crop
1 *harvests*

DC #2 - 11/15/2022: Oats, silage-soft dough

Field name: DC #2 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.83	25.22	305.27	2,356.14
Fresh water	0.00	0.00	0.00	704.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	236.83	25.22	305.27	3,061.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	226.69	49.46	362.70	1,676.66
Nutrient balance	10.14	-24.24	-57.43	1,384.35
Applied to removed ratio	1.04	0.51	0.84	1.83

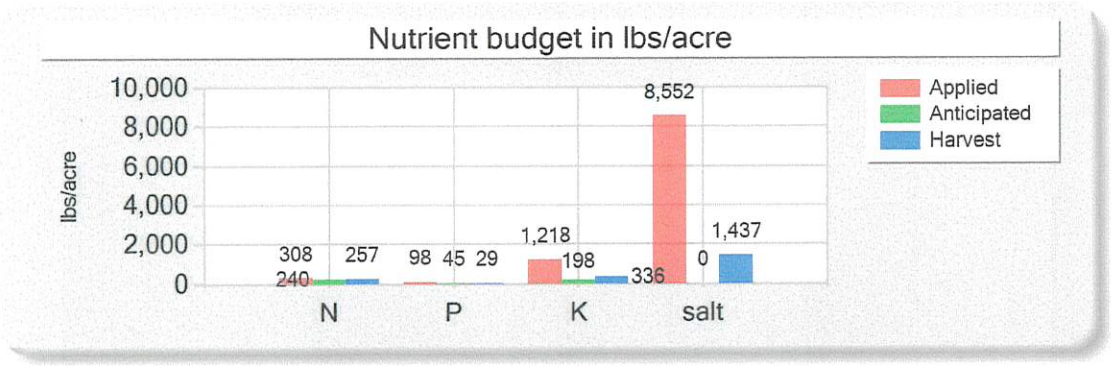
Fresh water applied
7,642,237.00 *gallons*
281.44 *acre-inches*
14.81 *inches/acre*

Process wastewater applied
924,000.00 *gallons*
34.03 *acre-inches*
1.79 *inches/acre*

Total harvests for the crop
1 *harvests*

DC #2 - 06/24/2023: Corn, silage

Field name: DC #2 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	301.22	97.63	1,218.46	6,794.79
Fresh water	0.00	0.00	0.00	1,757.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.22	97.63	1,218.46	8,552.07
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	256.65	28.52	336.50	1,437.25
Nutrient balance	51.57	69.12	881.96	7,114.83
Applied to removed ratio	1.20	3.42	3.62	5.95

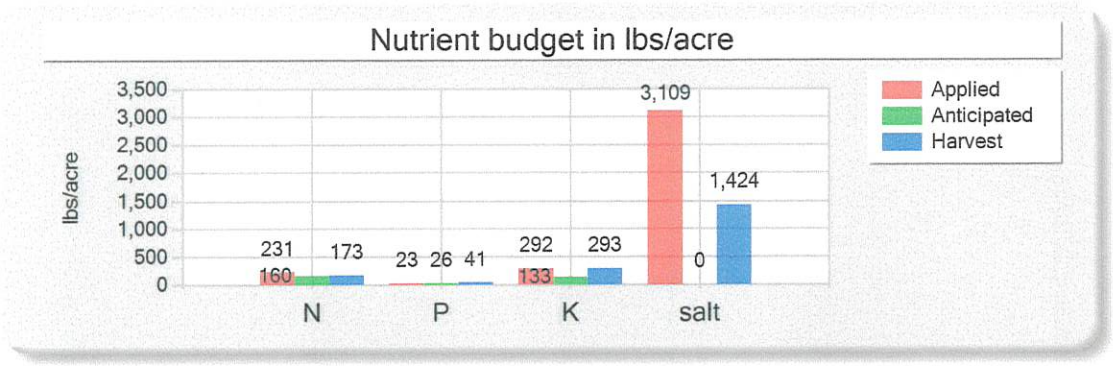
Fresh water applied
19,052,421.00 *gallons*
701.64 *acre-inches*
36.93 *inches/acre*

Process wastewater applied
4,059,000.00 *gallons*
149.48 *acre-inches*
7.87 *inches/acre*

Total harvests for the crop
1 *harvests*

DC #3 - 11/15/2022: Oats, silage-soft dough

Field name: DC #3 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.43	22.53	292.08	2,380.47
Fresh water	0.00	0.00	0.00	728.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.43	22.53	292.08	3,108.96
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	173.11	41.22	292.63	1,424.42
Nutrient balance	58.32	-18.69	-0.55	1,684.54
Applied to removed ratio	1.34	0.55	1.00	2.18

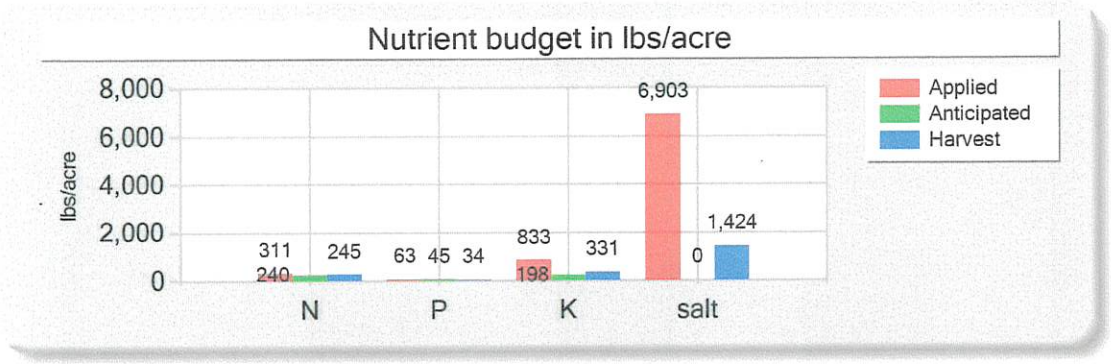
Fresh water applied
24,110,716.00 *gallons*
887.92 *acre-inches*
15.31 *inches/acre*

Process wastewater applied
2,751,000.00 *gallons*
101.31 *acre-inches*
1.75 *inches/acre*

Total harvests for the crop
1 *harvests*

DC #3 - 06/24/2023: Corn, silage

Field name: DC #3 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	304.12	62.51	833.27	5,101.78
Fresh water	0.00	0.00	0.00	1,801.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.12	62.51	833.27	6,903.11
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	245.25	34.22	330.80	1,423.57
Nutrient balance	65.87	28.29	502.47	5,479.54
Applied to removed ratio	1.27	1.83	2.52	4.85

Fresh water applied	
59,617,736.00	*gallons*
2,195.52	*acre-inches*
37.85	*inches/acre*

Process wastewater applied	
8,061,000.00	*gallons*
296.86	*acre-inches*
5.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Separated Solids

Sample and source description: 2022 Fall Separated Solids

Sample date: 11/15/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,000.00	10,600.00	17,300.00	30,700.00	11,800.00	10,500.00	8,700.00	2,350.00		42.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	2,000.00	7,600.00							69.60
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,300.00	23,500.00	6,900.00	2,900.00	3,300.00	2,300.00	591.00		55.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	490.00	342.00	0.00	0.00	28.80	570.00								6,550.00	6,780
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ1

Sample and source description: Imported WWSQ1

Sample date: 03/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	560.00	398.00	0.00	0.00	70.80	776.00								7,250.00	5,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	571.00	389.00	0.00	0.00	69.30	750.00								7,360.00	5,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ2

Sample and source description: Imported WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,200.00	535.00	0.00	0.00	55.60	889.00								7,840.00	7,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	689.00	378.00	0.00	0.00	42.50	850.00								8,170.00	7,360
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ3

Sample and source description: Imported WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	171.00	104.00	0.00	0.00	62.50	714.00	86.20	81.00	147.00	1,680.00	0.00	138.00	330.00	4,220.00	3,540
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	168.00	95.20	0.00	0.00	51.10	669.00	72.60	62.10	158.00	1,720.00	0.00	131.00	325.00	4,210.00	3,940
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	630.00	456.00	0.00	0.00	29.40	621.00								8,500.00	8,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

T.I.D Canal

Canal

Sample description: Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	210
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

DC #1 - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,100.00	7,400.00		13.10
DL	100.00	100.00	100.00		0.01

DC #1 - 06/24/2023: Corn, silage

DC #1 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	600.00	5,400.00		7.10
DL	100.00	100.00	100.00		0.01

DC #2 - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,200.00	8,800.00		11.30
DL	100.00	100.00	100.00		0.01

DC #2 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	500.00	5,900.00		7.50
DL	100.00	100.00	100.00		0.01

DC #3 - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,000.00	7,100.00		13.50
DL	100.00	100.00	100.00		0.01

DC #3 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	600.00	5,800.00		8.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

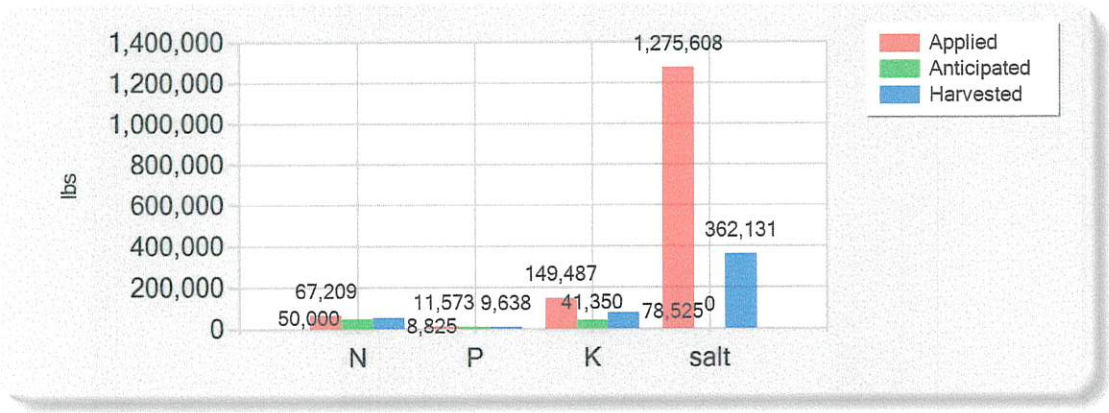
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

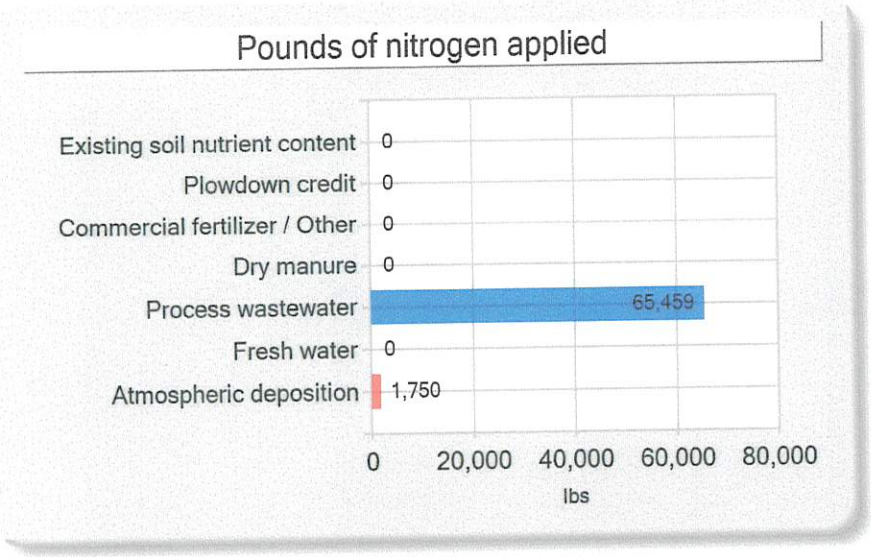
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

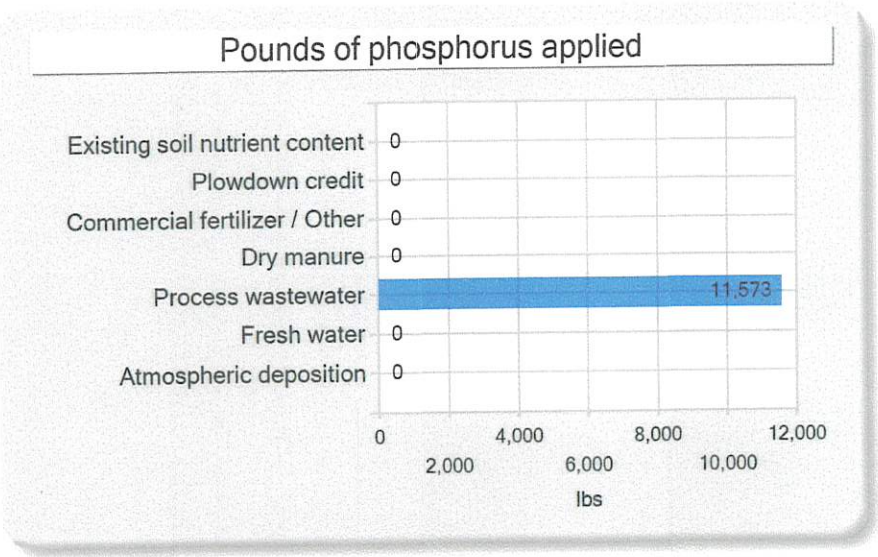
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	65,459.24	11,573.13	149,486.61	964,000.71
Fresh water	0.00	0.00	0.00	311,607.55
Atmospheric deposition	1,750.00	0.00	0.00	0.00
Total nutrients applied	67,209.24	11,573.13	149,486.61	1,275,608.26
Anticipated crop nutrient removal	50,000.00	8,825.00	41,350.00	0.00
Actual crop nutrient removal	55,235.32	9,637.68	78,525.36	362,130.94
Nutrient balance	11,973.92	1,935.45	70,961.26	913,477.32
Applied to removed ratio	1.22	1.20	1.90	3.52

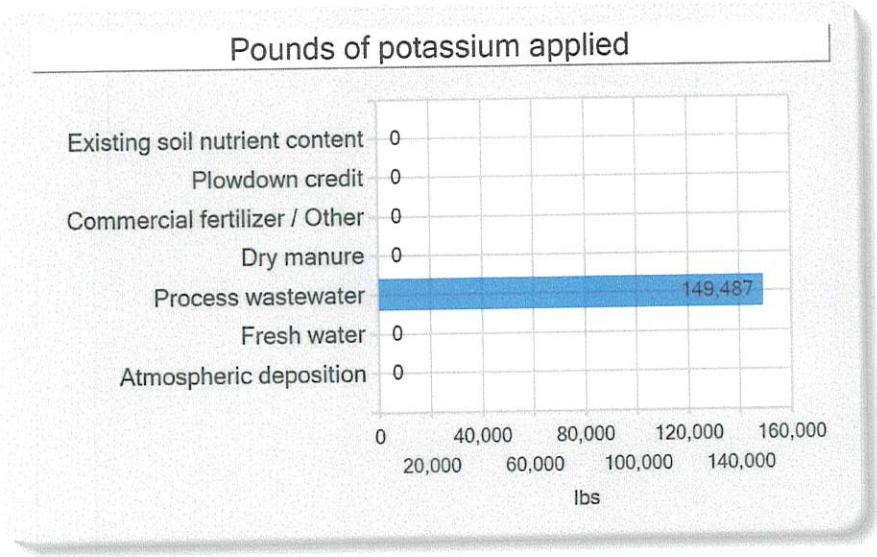
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

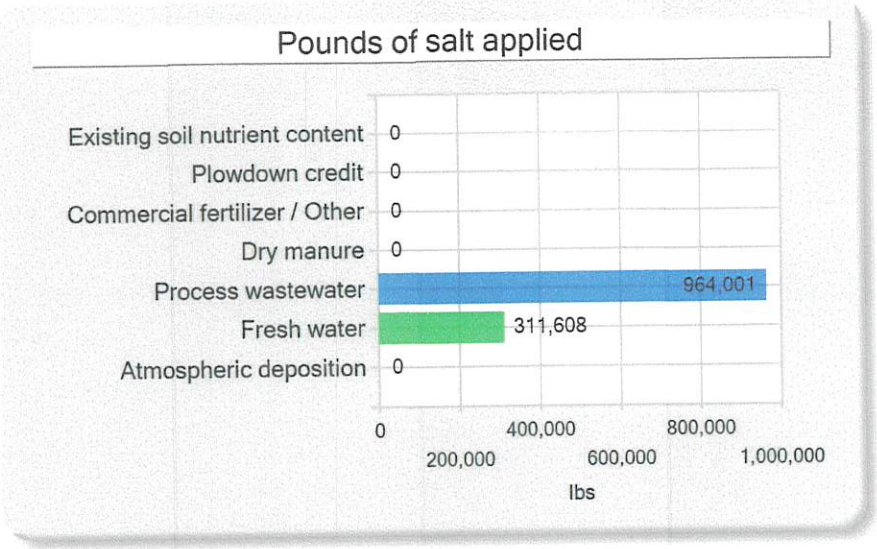


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No Milk Cows Onsite. Facility imports in wastewater from neighboring facility.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Gallie J Chaves	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
~~Dave Chaves~~ Galli Chaves	Mark Ahlem
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-26-24	6-17-24
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DC #4
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: SR
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	10/16	8:00	Domestic well	DW	15.0°C		✓				
2	↓	8:15	Irrigation well	DC4 Deep well	15.2°C						H7
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st		FRAGT				
2nd		DENELE ANALYTICAL, INC.	10/18/23 9:45			T3291009

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Double Cross #4
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW	T3291009A	Electrical Conductivity (EC)	EPA 120.1 and	1.86	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	46.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1190	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Double Cross #4
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DC4 Deep Well	T3291009B	Electrical Conductivity (EC)	EPA 120.1 and	1.51	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.3	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	775	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	27.7	mg/L
		Soluble Salts (SALT-SOL)	Calculation	963	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)			1.20
Electrical Conductivity (EC)	97.6			0.861 (T3291008A)	0.862 (Dup)	0.093
Nitrate Nitrogen (NO3-N)	96.9			23.4 (T3291008A)	22.8 (null)	2.49
Total Dissolved Solids (TDS)	93.5			1190 (T3296001E)	1150 (Dup)	3.43
Total Nitrogen (TN)	98.6			8.40 (T3291009B)	8.40 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714